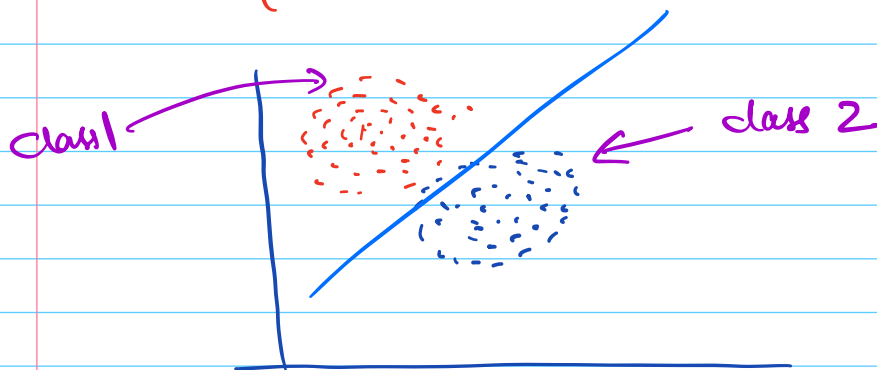


Lec. 14

Machine Learning.
(ML).

classification with
Gaussian classes



classes are Gaussian

$$\text{class 1 : } N(\bar{\mu}_1, \Sigma)$$

$$\text{class 2 : } N(\bar{\mu}_2, \Sigma)$$

New observation : $\bar{x}$

which class does $\bar{x}$
belong to ?

Look at the probability density for
each class.

$$p(\bar{x}; \bar{\mu}_1, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \times$$

↓
'likelihood'
of class 1.

$$e^{-\frac{1}{2}(\bar{x}-\bar{\mu}_1)^T \Sigma^{-1}(\bar{x}-\bar{\mu}_1)}$$

← Determinant

$$P(\bar{x}; \bar{\mu}_2, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(\bar{x} - \bar{\mu}_2)^T \Sigma^{-1} (\bar{x} - \bar{\mu}_2)}$$

↓
Likelihood of
class 2.

Classifier: Likelihood class 1 $>$ Likelihood class 2

$\Rightarrow$ choose class 1 !

else choose class 2

$$\underbrace{P(\bar{x}; \bar{\mu}_1, \Sigma)}_{\text{likelihood } C_1} \underset{C_2}{\overset{C_1}{\geq}} \underbrace{P(\bar{x}; \bar{\mu}_2, \Sigma)}_{\text{likelihood } C_2}$$

Choose C_1 if:

$$\frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(\bar{x} - \bar{\mu}_1)^T \Sigma^{-1} (\bar{x} - \bar{\mu}_1)} \geq$$

$$\frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(\bar{x} - \bar{\mu}_2)^T \Sigma^{-1} (\bar{x} - \bar{\mu}_2)}$$

choose C_1 if

$$\text{i.e. } (\bar{x} - \bar{\mu}_1)^T \Sigma^{-1} (\bar{x} - \bar{\mu}_1) \leq (\bar{x} - \bar{\mu}_2)^T \Sigma^{-1} (\bar{x} - \bar{\mu}_2)$$

$$\Rightarrow \cancel{\bar{x}^T \Sigma^{-1} \bar{x}} - \bar{\mu}_1^T \Sigma^{-1} \bar{x} - \bar{x} \Sigma^{-1} \bar{\mu}_1 + \bar{\mu}_1^T \Sigma^{-1} \bar{\mu}_1$$

$$\leq \cancel{\bar{x}^T \Sigma^{-1} \bar{x}} - \bar{\mu}_2^T \Sigma^{-1} \bar{x} - \bar{x}^T \Sigma^{-1} \bar{\mu}_2 + \bar{\mu}_2^T \Sigma^{-1} \bar{\mu}_2$$

$$\Rightarrow (\bar{\mu}_2 - \bar{\mu}_1)^T \Sigma^{-1} \bar{x} \leq \frac{1}{2} \bar{\mu}_2^T \Sigma^{-1} \bar{\mu}_2 - \frac{1}{2} \bar{\mu}_1^T \Sigma^{-1} \bar{\mu}_1$$

$$\Rightarrow \underbrace{(\bar{\mu}_1 - \bar{\mu}_2)^T \Sigma^{-1} \left(\bar{x} - \frac{\bar{\mu}_2 + \bar{\mu}_1}{2} \right)}_{\text{choose } C_1} \geq 0.$$

$$\Rightarrow \boxed{\bar{w}^T (\bar{x} - \bar{x}_0) \underset{C_2}{\overset{C_1}{\geq}} 0}$$

where $\bar{w} = \Sigma^{-1} (\bar{\mu}_1 - \bar{\mu}_2)$

$\underbrace{\bar{w}^T \bar{x} = \bar{w}^T \bar{x}_0}_{\substack{\uparrow \\ \text{hyperplane}}}$

$$\bar{x}_0 = \frac{\bar{\mu}_1 + \bar{\mu}_2}{2}$$

$$\Sigma = \underbrace{\sigma^2 \mathbf{I}}_{\text{for i.i.d Gaussian}}$$

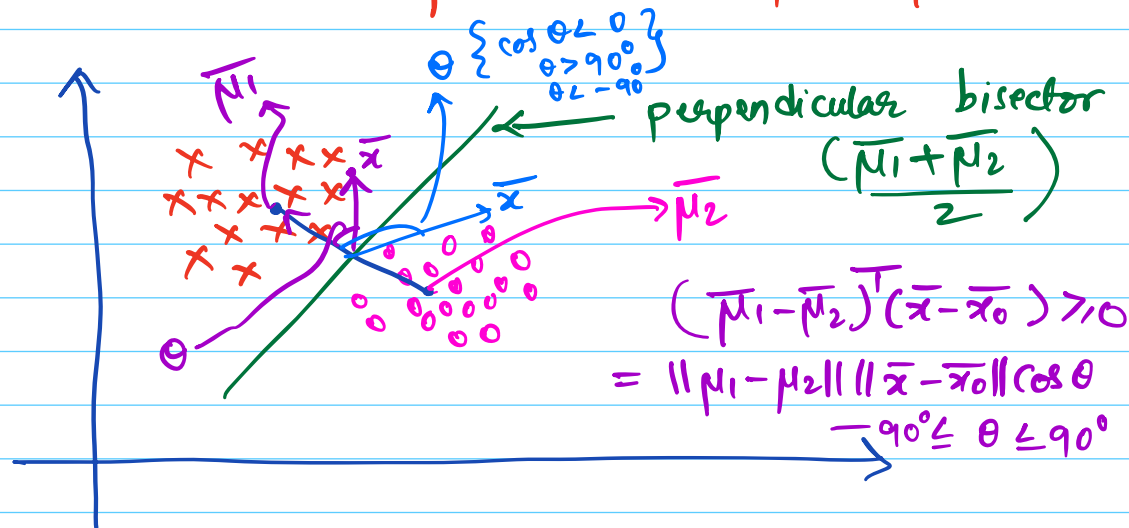
$$\bar{w} = \frac{1}{\sigma^2} (\bar{\mu}_1 - \bar{\mu}_2)$$

$$\frac{1}{\sigma^2} (\bar{\mu}_1 - \bar{\mu}_2)^T (\bar{x} - \bar{x}_0) \underset{C_2}{\overset{C_1}{\geq}} 0$$

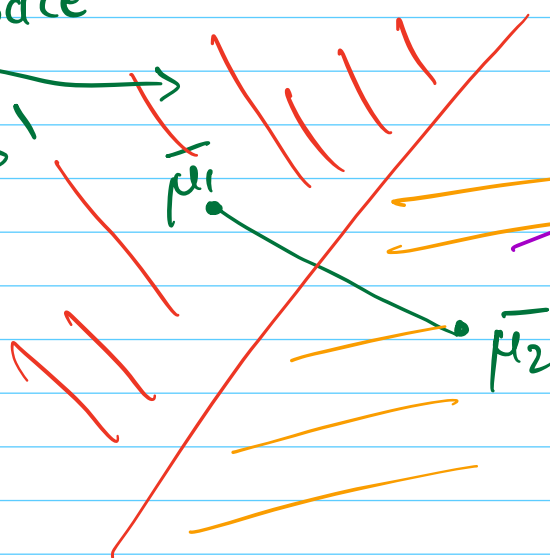
$$(\bar{\mu}_1 - \bar{\mu}_2)^T (\bar{x} - \bar{x}_0) \begin{matrix} \geq c_1 \\ \leq c_2 \end{matrix} 0$$

$$\bar{x}_0 = \frac{\bar{\mu}_1 + \bar{\mu}_2}{2}$$

mid-point of $\bar{\mu}_1, \bar{\mu}_2$



half space
classified as class 1



classified as class 2.

$$\|\bar{x} - \bar{\mu}_1\| \begin{matrix} \leq c_1 \\ \geq c_2 \end{matrix} \|\bar{x} - \bar{\mu}_2\|$$

minimum distance classifier!

